Mac Lee, PhD

(626) 802-0859 | macthecadillac@gmail.com | GitHub: macthecadillac | US citizen

Physics PhD; applied ML; statistics & optimization; high-performance computing; author of a production open-source library (11.3M+ downloads); co-maintainer of a large-scale open-source ML framework

Education

University of California, San Diego **October 2025**

PhD in Physics, Specializing in Physical Oceanography La Jolla, CA

- Worked on extended surface quasigeostrophy (SQG+), a reduced model for the submesoscale upper ocean, using asymptotic analysis. The model is a candidate theory for the parametrization of submesoscale physics which cannot yet be directly resolved by modern supercomputers in large-scale ocean and climate simulations.
- Worked on the trajectories and statistics of the Lagrangian particles for a special case of SQG+ using analytical and numerical methods. Discovered conservations laws for this special case of SQG+ flow.
- Worked on the flow stability of SQG+ using analytical and numerical methods. Created different phase diagrams for the flow regime. Used Python, Rust and MPI heavily in high performance simulations.
- Optimized numerical convergence behavior using series acceleration methods.
- Relevant courses: PHYS 239 Nonlinear Dynamics and Mathematical Modeling; CSE 230 Principles of Programming Languages; CSE 291 Program Synthesis; CSE 231 Advanced Compiler Design; CSE 291 Browser and Wasm Security

California State University, Northridge **May 2018**

M.S. in Physics with Distinction, Specializing in Condensed Matter Theory Northridge, CA

- Worked on quantum decoherence, topological superconductors, and quantum spin liquids
- Wrote high performance Python code for large scale numerical simulations. Optimized Hamiltonian generator function to achieve a 20x runtime reduction (reduced from 1 hour to 3 minutes per call). Optimized memory usage using quantum number conservation (over 250x reduction in memory use). Scientific computing on CPU clusters.
- Analyzed data from collaborators and performed visualization using linear regression and Fourier analysis

Experience

FastVideo (open-source contributions) **May 2026 – present**

Open-source inference and post-training framework for accelerated video generation; 3.5k+ GitHub stars, adopted/forked by SGLang for multimodal inference, integrated as a backend in NVIDIA Dynamo, and used by Tencent HY-WorldPlay. https://github.com/hao-ai-lab/FastVideo

- Collaborator / co-maintainer since June 2026
- Co-authored commits to port FLUX.2, a text-to-image flow-matching model, into FastVideo
- Ongoing work with a reinforcement learning training and inference pipeline, a performance benchmark regression tracking redesign, and Trajectory Distribution Matching, a distillation framework

University of California, San Diego **March 2020 – September 2024**

Associate / Senior TA / Lab Teaching Assistant Coordinator La Jolla, CA

- Supervised and trained lab teaching assistants and graders; held office hours; contributed to full curriculum rewrite

Skills

- Physical simulation; Dynamical systems; Fluid mechanics; Fourier analysis; Asymptotic analysis; Lagrangian systems; Series acceleration; Data visualization
- 10 years of experience with Python, Rust, Numpy, SciPy, Matplotlib, git
- Other technical skills: PyTorch; Pandas; Sympy; Julia; Haskell; OCaml; TypeScript; LLVM IR; WebAssembly; MPI; Docker

Publications

- M. Lee and S. G. Llewellyn Smith, “SQG point vortex dynamics with order Rossby corrections”, *European Journal of Mechanics B Fluids* 118, 204467 (2026).
- M. Lee and S. G. Llewellyn Smith, “Stability of SQG Kolmogorov Flow”, *Dynamics of Atmospheres and Oceans*, 101680 (2026).
- S.-S. Gong, W. Zheng, M. Lee, Y.-M. Lu, and D. N. Sheng, “Chiral spin liquid with spinon Fermi surfaces in the spin-1/2 triangular Heisenberg model”, *Physical Review B* 100, 241111 (2019).
- M. Lee, T. R. Look, S. P. Lim, and D. N. Sheng, “Many-body localization in spin chain systems with quasiperiodic fields”, *Physical Review B* 96, 075146 (2017).

Talks and Presentations

- “SQG point vortex dynamics with order Rossby corrections”, SoCal Fluids XIX (University of California, Riverside, CA, Apr. 18, 2026), Talk.
- “SQG point vortex dynamics with order Rossby corrections”, Workshop on Numerical and Theoretical Advances in Geophysical Fluid Dynamics and Nonlinear PDE Systems (Texas A&M University, College Station, TX, Feb. 7, 2026), Talk.
- “Stability of SQG Kolmogorov flow with order Rossby corrections”, SoCal Fluids XVII (University of California, Irvine, CA, Apr. 13, 2024), Talk.
- “Spin Liquid and Quantum Phase Diagram on a Spin-Orbit Coupled Triangular Lattice”, Inter- national Conference on Magnetism (San Francisco, CA, July 16, 2018), Poster Presentation.
- “Spin Liquid and Quantum Phase Diagram on a Spin-Orbit Coupled Triangular Lattice”, APS March Meeting (Los Angeles, CA, Mar. 7, 2018), Talk.
- “Topological Insulators”, Physics Symposium (California State University, Northridge, CA, July 2017), Poster Presentation.
- “Many-Body Localization in Spin Chain Systems with Quasiperiodic Fields”, CSUNposium (California State University, Northridge, CA, Apr. 7, 2017), Talk.

Projects

- Project: https://crates.io/crates/atree
  - An open-source high performance, arena-based tree structure with node removal written in Rust
  - Published on crates.io, downloaded over 11.3 million times, and used by C2PA, an Adobe-backed non-profit organization.
- Project: https://github.com/macthecadillac/TFNet
  - Reproduction of Wang et al., “Towards physics-informed deep learning for turbulent flow prediction”, 2020.
  - Physics-informed flow prediction without architectural guarantees of flow invariants
- Project: https://github.com/macthecadillac/spike-train-autoencoder
  - Spike-Train Autoencoder — Spiking Neural Network for Image Encoding
  - A proof-of-concept image autoencoder using leaky integrate-and-fire (LIF) neuron layers, exploring how spiking neural networks can serve as biologically-grounded feature encoders
  - Encoder, decoder, and adapter trained on the STL-10 dataset with embeddings aligned to OpenAI CLIP representations

Awards, Grants and Honors

- 2024 Office of Naval Research Grant (University of California, San Diego)
- 2018 Mack I. Johnson Research Award for Outstanding Graduate Student in the College of Science and Mathematics (California State University, Northridge)
- 2018 C. Y. Liang Outstanding Graduate Student Award (California State University, Northridge)
- 2017 Leslie and Terry Cutler Scholarship (California State University, Northridge)